Practice workbook · Describe genetic modification and gene therapy for human health

Engineering bacteria or yeast to make human insulin, Golden Rice for vitamin A, and gene therapy that delivers a working allele into body cells with a viral vector.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-GENETIC-MODIFICATION

1. Question 6 · part whole · 6 marks

June 2022 Paper 1

Worked-example retry · Attempt part whole only.

6 Describe how bacteria can be genetically modified to produce human insulin.

(6)

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(Total for Question 6 = 6 marks)

1. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

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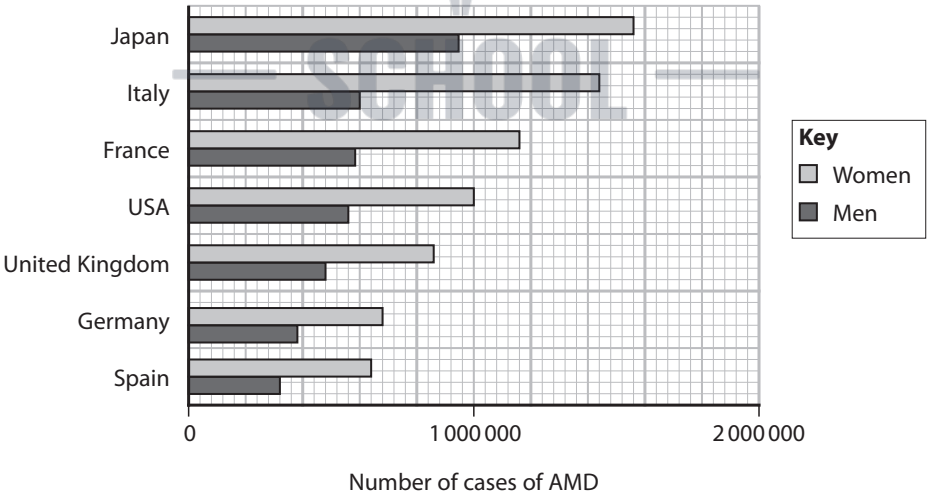
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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Turn over ▶

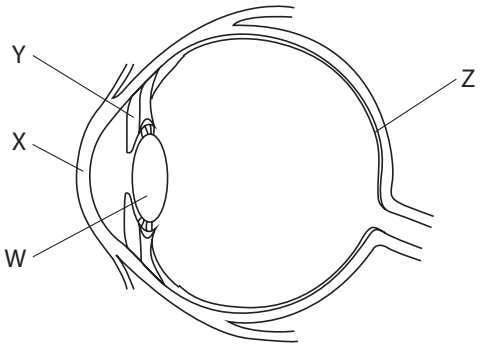
2. Question 6 · part c · 3 marks

January 2023 Paper 2

Further practice · Attempt part c only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.



Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

(Total for Question 6 = 11 marks)

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2. Question 6 · part c · 3 marks · Working and answer

Working and answer

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3. Question 7 · part e · 2 marks

June 2025 Paper 2R

Further practice · Attempt part e only.

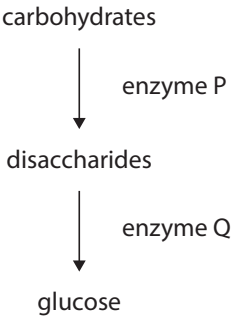
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

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(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

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(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

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3. Question 7 · part e · 2 marks

June 2025 Paper 2R

Further practice · Attempt part e only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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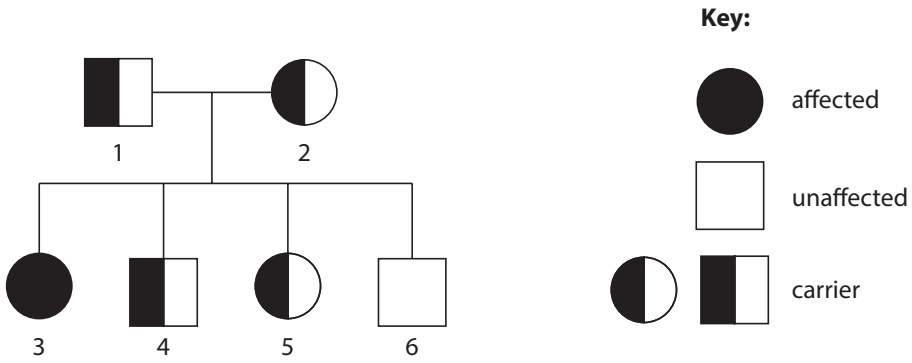
Working and answer

4. Question 8 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

8 (a) The diagram shows the inheritance of cystic fibrosis in a family.



(Source: Adapted from https://cysticfibrosiszaw2.weebly.com/)

(i) Which individual is homozygous recessive?

(1)

- ☒ **A** 1
- ☒ **B** 3
- ☒ **C** 5
- ☒ **D** 6

(ii) How many individuals have a heterozygous genotype?

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(iii) Explain why carriers of cystic fibrosis do not have symptoms of the disease.

(2)

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P 7 4 6 3 0 A 0 2 2 2 4

4. Question 8 · part b · 3 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(iv) Draw a genetic diagram to show how two parents can have a 1 in 4 chance of having a child with cystic fibrosis.

Use F for the non-cystic fibrosis allele and f for the cystic fibrosis allele.

(4)

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(b) Describe how gene therapy can be used to treat cystic fibrosis.

(3)

(Total for Question 8 = 11 marks)

4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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